# MTH316 2021S1 Exam

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* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
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* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference
* MTH355 Basic Mathematical Optimisation

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# MTH316

**Timed Online Assignment – January Semester 2021**

## Multivariable Calculus

**Friday, 21 May 2021** **4:00 pm – 6:30 pm**

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**Time allowed: 2.5 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O)*.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

*Answer all questions. (Total 100 marks)*

**Question 1**

A function of two variables $(x, y)$ is given as:

$$f(x,y) = \frac{x^3 + y^3}{\sqrt{x^3 + y^3 + 1} - 1}$$

(a) Determine the existence of $\lim_{(x,y)\to(1,1)} f(x,y)$. If exist, evaluate it.

(4 marks)

(b) Determine the existence of $\lim_{(x,y)\to(0,0)} f(x,y)$. If exist, evaluate it.

(6 marks)

(c) If there is a definition at point $(0,0)$, $f(x,y) = m$. Find the value of $m$ to make the function $f(x,y)$ is continuous at point $(0,0)$.

(4 marks)

**Question 2**

Suppose near a cargo ship, the depth of the sea is given by the function of:

$$f(x,y) = 500 - 0.02x^2 - 0.03y^2,$$

where $x$, $y$ are measured in meters. Suppose a fisherman is on a small boat with coordinates (30, 40, 434). The positive $x$ -axis points east and the positive y-axis points north.

(a) Calculate the gradient of $f(x,y)$ at (30, 40, 434).

(4 marks)

(b) If the fisherman move towards North, is the water under the boat getting shallower or deeper when he departs?

(4 marks)

(c) If the fisherman move towards to the cargo ship, will he start to ascend or descend? At what rate?

(4 marks)

(d) If the fisherman starts to moves, in which direction is the water under the boat getting deepest? What is the rate of change of the depth in that direction?

(4 marks)

**Question 3**

Skip.

(a) Apply Lagrange Multipliers to find the maximum and minimum of the function

Skip

$$f(x, y, z) = x^2 + 2xy - 2yz - z^2$$

on the unit sphere $x^2 + y^2 + z^2 = 1$.

(10 marks)

(b) Let five numbers $e_1$, $e_2$, $e_3$, $e_4$, $e_5$ be given and $f(x) = ax^2 + bx + c$. Determine the values of $a$, $b$ and $c$ that minimizes the total square error

$$E = [f(-2) - e_1]^2 + [f(-1) - e_2]^2 + [f(0) - e_3]^2 + [f(1) - e_4]^2 + [f(2) - e_5]^2.$$

(10 marks)

**Question 4**

A solid $E$ lies between the parabolic surface $z = 4 - x^2$ and three planes $x = 0$, $y = 0$ and $y = x + 2$ as seen in Figure Q4.

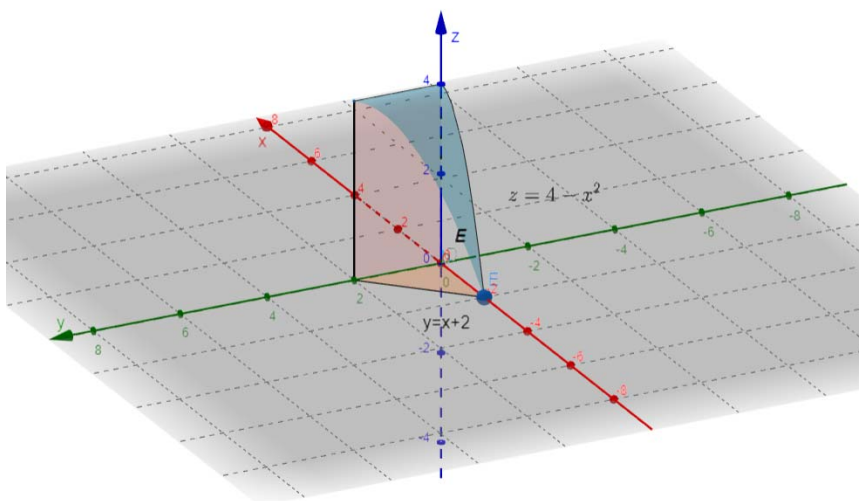



**Figure Q4**

(a) Find a triple integral to express the volume of the solid E, integrating first with respect to $z$, then $y$, and then $x$.

(6 marks)

(b) Calculate the volume of solid E.

(6 marks)

**Question 5**

Define the vector field **F** by

$$\mathbf{F} = (x^2 - \mathrm{y})\,\mathbf{i} + (x - xy)\mathbf{j}.$$

Let $C$ be the curve consisting of $C_1$, $C_2$ and $C_3$ as shown in the following Figure Q5. $C_1$ is a line from $A(0,0)$ to $B(0,1)$, $C_2$ is a quarter circle with the parameter equation $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$ $(\pi/2 \leqslant t \leqslant \pi)$. $C_3$ is a line from $C(-1,0)$ to $A(0,0)$.

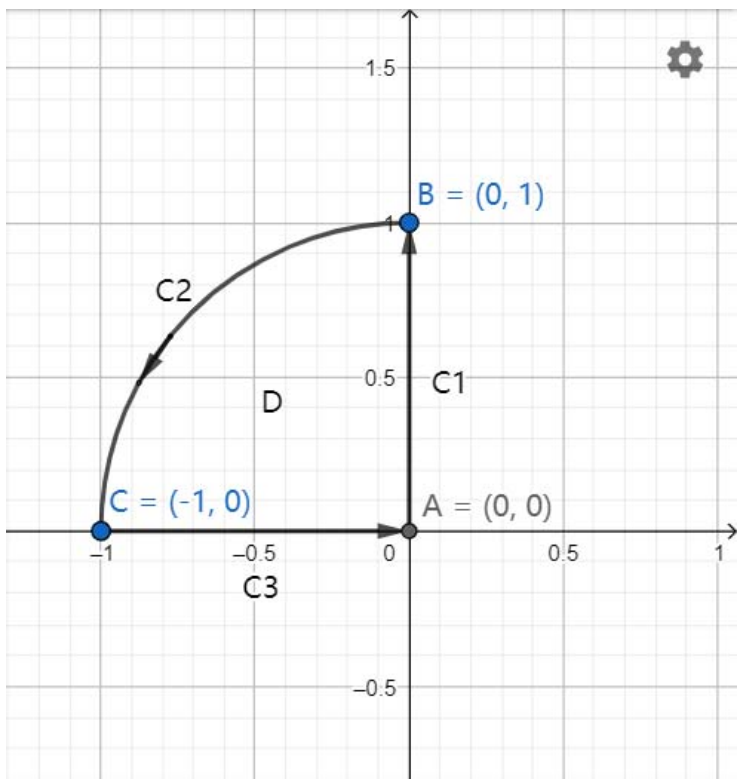



**Figure Q5**

(a) Compute the line integral of **F** along $C_1, C_2, C_3$, $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$, $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ and $\int_{C_3} \mathbf{F} \cdot d\mathbf{r}$.

(9 marks)

(b) Find the curl of **F**.

(4 marks)

(c) Suppose D is enclosed by $C_1, C_2$ and $C_3$. Use Green's Theorem to evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

(6 marks)

**Question 6**

Define the vector field **F** by

$$\mathbf{F}(x, y, z) = -xy\mathbf{i} + y\mathbf{j} + \mathrm{yz}\mathbf{k}.$$

Let $C$ be the line segment from $(0,1,0)$ to $(1,0,0)$, from $(1,0,0)$ to $(0,0,1)$, and from $(1,0,0)$ back to $(0,1,0)$ as seen in Figure Q6.

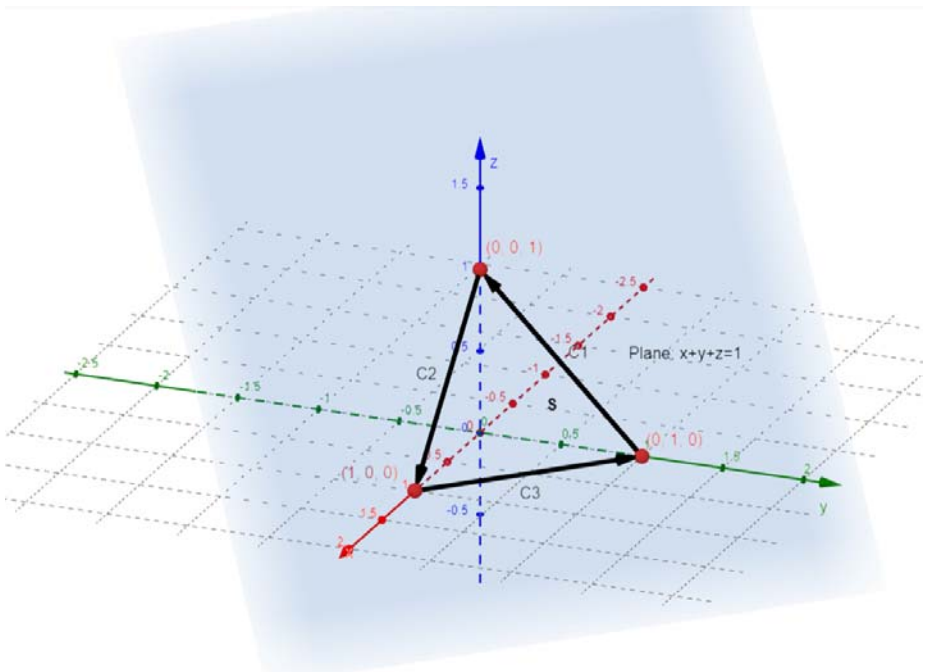



**Figure Q6**

(a) Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

(9 marks)

(b) Evaluate curl **F**.

(4 marks)

(c) Suppose $\boldsymbol{S}$ is the part of the plane $x + y + z = 1$ that lies in the first octant, oriented outward. Use Stokes' Theorem to compute the integral $\iint_S \mathrm{curl}\,\mathbf{F} \cdot d\mathbf{S}$

(6 marks)

**----- END OF PAPER -----**

1(a) $f(x,y) = \dfrac{x^3+y^3}{\sqrt{x^3+y^3+1}-1}$

$$\lim_{(x,y)\to(1,1)} f(x,y) = f(1,1)$$

$$= \frac{1+1}{\sqrt{1+1+1}-1}$$

$$= \frac{2}{\sqrt{3}-1}$$

(b) $$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to 0} \frac{x^3+y^3}{\sqrt{x^3+y^3+1}-1}$$

$$= \lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{\sqrt{x^3+y^3+1}-1} \cdot \frac{\sqrt{x^3+y^3+1}+1}{\sqrt{x^3+y^3+1}+1}$$

$$= \lim_{(x,y)\to(0,0)} \frac{(x^3+y^3)\left[\sqrt{x^3+y^3+1}+1\right]}{(x^3+y^3+1)-1}$$

$$= \lim_{(x,y)\to(0,0)} \frac{(x^3+y^3)\left[\sqrt{x^3+y^3+1}+1\right]}{x^3+y^3}$$

$$= \lim_{(x,y)\to(0,0)} \left[\sqrt{x^3+y^3+1}+1\right]$$

$$= \lim_{(x,y)\to(0,0)} \left[\sqrt{0+0+1}+1\right]$$

$$= 2$$

(c)

$$\therefore f(x,y) = \begin{cases} \dfrac{x^3+y^3}{\sqrt{x^3+y^3+1}-1} & \text{if } (x,y) \neq (0,0) \\ 2 & \text{if } (x,y) = (0,0) \end{cases}$$

since $f(0,0) = \lim_{(x,y)\to(0,0)} f(x,y)$, then $f(x,y)$ is continuous

2(a) $f(x,y) = 500 - 0.02x^2 - 0.03y^2$

$f_x(x,y) = -0.04x$

$f_y(x,y) = -0.06y$

gradient at $(30, 40, 434)$ $= \nabla f(30,40)$

$$= \begin{bmatrix} f_x(30,40) \\ f_y(30,40) \end{bmatrix}$$

$$= \begin{bmatrix} -0.04(30) \\ -0.06(40) \end{bmatrix}$$

$$= \begin{bmatrix} -1.2 \\ -2.4 \end{bmatrix}$$

(b) $\underset{\sim}{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$D_u f(30,40) = \nabla f(30,40) \cdot \underset{\sim}{u}$$

$$= \begin{bmatrix} -1.2 \\ -2.4 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= -2.4$$

$\therefore$ The water depth decreases

(c) Suppose the cargo boat is at $(0,0)$

$\underset{\sim}{p} = \begin{bmatrix} 30 \\ 40 \end{bmatrix}$ $\quad |\underset{\sim}{p}| = \sqrt{30^2+40^2} = \sqrt{50^2} = 50$

let $\underset{\sim}{u} = \frac{1}{50}\begin{bmatrix} 30 \\ 40 \end{bmatrix}$

$$D_u f(30,40) = \nabla f(30,40) \cdot \underset{\sim}{u}$$

$$= \begin{bmatrix} -1.2 \\ -2.4 \end{bmatrix} \cdot \frac{1}{50}\begin{bmatrix} 30 \\ 40 \end{bmatrix}$$

$$= -\frac{132}{50}$$

The depth of the ocean decreases at rate of $\frac{-132}{50}$ m per m moved by the fisherman as he moves towards the cargo ship.

2(d) direction for max depth

$$= \frac{\nabla f(30,40)}{|\nabla f(30,40)|}$$

$$= \frac{1}{\sqrt{(-1.2)^2 + (2.4)^2}} \begin{bmatrix} -1.2 \\ -2.4 \end{bmatrix}$$

$$= \frac{1}{7.2} \begin{bmatrix} -1.2 \\ -2.4 \end{bmatrix}$$

rate of change $= |\nabla f(30,40)|$

$= 7.2$

3(a) Lagrange Multiplier not examinable for 2024S1. Skipped.

3(b) $E = (f(-2)-e_1)^2 + [f(-1)-e_2]^2 + [f(0)-e_3]^2 + [f(1)-e_4]^2 + [f(2)-e_5]^2$

$= [4a-2b+c-e_1]^2 + [a-b+c-e_2]^2 + [a+b+c-e_4]^2 + [4a+2b+c-e_5]^2$

$\frac{\partial E}{\partial a} = 2(4a-2b+c-e_1)(4) + 2(a-b+c-e_2)(1) + 2(a+b+c-e_4)(1) + 2(4a+2b+c-e_5)(4)$

$= 68a + 20c - 10e_1 - 2e_4 - 8e_3$

$\frac{\partial E}{\partial b} = 2(4a-2b+c-e_1)(-2) + 2(a-b+c-e_2)(-1) + 2(a+b+c-e_4)(1)$
$+ 2(4a+2b+c-e_3)(2)$

$= 20b + 6e_1 - 2e_4 - 4e_3$

$\frac{\partial E}{\partial c} = 2(4a-2b+c-e_1)(1) + 2(a-b+c-e_2)(1) + 2(a+b+c-e_4) + 2(4a+2b+c-e_5)(1)$

$= 68a + 20c - 10e_1 - 2e_4 - 8e_3$

$= 20a + 10c - 4e_1 - 2e_3 - 2e_4 - 2e_5$

let $\frac{\partial E}{\partial a} = \frac{\partial E}{\partial b} = \frac{\partial E}{\partial c} = 0$

$20b + 6e_1 - 2e_4 - 4e_5 = 0$

$\therefore b = \frac{1}{20}[-6e_1 + 2e_4 + 4e_5]$

$= \frac{1}{10}[-3e_1 + e_4 + 2e_5]$

$68a + 20c = 10e_1 + 2e_4 + 8e_3 \quad -(1)$

$20a + 10c = 4e_1 + 2e_3 + 2e_4 + 2e_5 \quad -(2)$

(1)-2(2): $28a = 2e_1 - 4e_3 - 2e_4 + 4e_5$

$\therefore a = \frac{1}{28}[2e_1 - 4e_3 - 2e_4 + 4e_5]$

$= \frac{1}{14}[e_1 - 2e_3 - e_4 + 2e_5]$

$10c = 4e_1 + 2e_3 + 2e_4 + 2e_5 - 20a$

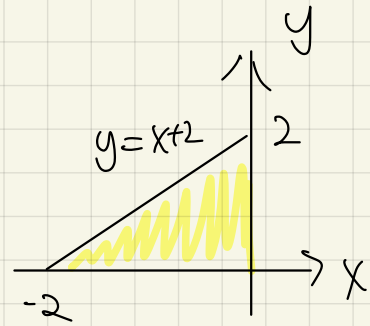
$= \frac{18}{7}e_1 + \frac{34}{7}e_3 + \frac{24}{7}e_4 - \frac{6e_5}{7}$

$c = \frac{9}{35}e_1 + \frac{17}{35}e_3 + \frac{12}{35}e_4 - \frac{3}{35}e_5$

4(a) $0 \le z \le 4-x^2$

$0 \le y \le x+2$

$-2 \le x \le 0$




$$\therefore \text{volume} = \int_{-2}^{0}\int_{0}^{x+2}\int_{0}^{4-x^2} 1 \, dz\, dy\, dx$$

(b) $$\text{volume} = \int_{-2}^{0}\int_{0}^{x+2}\int_{0}^{4-x^2} 1 \, dz\, dy\, dx$$

$$= \int_{-2}^{0}\int_{0}^{x+2} 4-x^2 \, dy\, dx$$

$$= \int_{-2}^{0} (4-x^2)(x+2)\, dx$$

$$= \int_{-2}^{0} -x^3 - 2x^2 + 4x + 8 \, dx$$

$$= \left[-\tfrac{1}{4}x^4 - \tfrac{2}{3}x^3 + 2x^2 + 18x\right]_{-2}^{0}$$

$$= \frac{20}{3}$$

5(a) $F = \begin{bmatrix} x-y \\ x-xy \end{bmatrix}$

$C_1$: $r(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ t \end{bmatrix}$ $0 \le t \le 1$

$r'(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$F(t) = F(0,t)$
$= \begin{bmatrix} -t \\ 0 \end{bmatrix}$

$\int_{C_1} F \cdot dr = \int_0^1 F(t) \cdot r'(t)\, dt$
$= \int_0^1 \begin{bmatrix} -t \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} dt$
$= \int_0^1 0\, dt$
$= 0$

$C_2$: $r(t) = \begin{bmatrix} \cos(t) \\ \sin(t) \end{bmatrix}$ for $\frac{\pi}{2} \le t \le \pi$

$r'(t) = \begin{bmatrix} -\sin(t) \\ \cos(t) \end{bmatrix}$

$F(t) = F(\cos(t), \sin(t)) = \begin{bmatrix} \cos^2 t - \sin t \\ \cos t - \cos t \sin t \end{bmatrix}$

$\int_{C_2} F \cdot dr = \int_{\frac{\pi}{2}}^{\pi} F(t) \cdot r'(t)\, dt$
$= \int_{\frac{\pi}{2}}^{\pi} \begin{bmatrix} \cos^2(t) - \sin(t) \\ \cos(t) - \cos(t)\sin(t) \end{bmatrix} \cdot \begin{bmatrix} -\sin(t) \\ \cos(t) \end{bmatrix} dt$
$= \int_{\frac{\pi}{2}}^{\pi} \sin^2(t) - \sin(t)\cos^2(t) + \cos^2(t) - \sin(t)\cos^2(t)\, dt$
$= \int_{\frac{\pi}{2}}^{\pi} 1 - 2\sin(t)\cos^2(t)\, dt$
$= \left[t + \frac{2}{3}\cos^3(t)\right]_{\frac{\pi}{2}}^{\pi}$
$= \left(\pi - \frac{2}{3}\right) - \left(\frac{\pi}{2} + 0\right)$
$= \frac{\pi}{2} - \frac{2}{3}$

$C_3$: $r(t) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} t-1 \\ 0 \end{bmatrix}$ for $0 \le t \le 1$

$r'(t) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$F(t) = F(t-1, 0)$
$= \begin{bmatrix} (t-1)^2 \\ t-1 \end{bmatrix}$

$\int_{C_1} F \cdot dr = \int_0^1 F(t) \cdot r'(t)\, dt$
$= \int_0^1 \begin{bmatrix} (t-1)^2 \\ t-1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} dt$
$= \int_0^1 (t-1)^2\, dt$
$= \left[\frac{1}{3}(t-1)^3\right]_0^1$
$= \frac{1}{3}$

5(b) $\text{curl}\,F = \nabla \times F$

$$= \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y & x - xy & 0 \end{vmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ \frac{\partial}{\partial x}(x - xy) - \frac{\partial}{\partial y}(x^2 - y) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ (1-y) - (-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2-y \end{bmatrix}$$

(c) $C_1 + C_2 + C_3$ is in anti-clockwise direction

$D$: $r(\rho, \theta) = \begin{bmatrix} \rho\cos\theta \\ \rho\sin\theta \end{bmatrix}$ where $0 \le \rho \le 1$, $\frac{\pi}{2} \le \theta \le \pi$

$\frac{\partial Q}{\partial x} = 1 - y \qquad \frac{\partial P}{\partial y} = -1$

By Green's Theorem,

$$\int_{C_1+C_2+C_3} F \cdot dr = \iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dA$$

$$= \iint_D (1-y) - (-1)\, dA$$

$$= \iint_D 2 - y\, dA$$

$$= \int_{\frac{\pi}{2}}^{\pi} \int_0^1 (2 - \rho\sin\theta)\rho\, d\rho\, d\theta$$

$$= \int_{\frac{\pi}{2}}^{\pi} \int_0^1 2\rho - \rho^2 \sin\theta\, d\rho\, d\theta$$

$$= \int_{\frac{\pi}{2}}^{\pi} \left[ \rho^2 - \frac{1}{3}\rho^3 \sin\theta \right]_{\rho=0}^{\rho=1} d\theta$$

$$= \int_{\frac{\pi}{2}}^{\pi} 1 - \frac{1}{3}\sin\theta\, d\theta$$

$$= \left[ \theta + \frac{1}{3}\cos\theta \right]_{\theta=\frac{\pi}{2}}^{\theta=\pi}$$

$$= \left(\pi + \frac{1}{3}\cos\pi\right) - \left(\frac{\pi}{2} + \frac{1}{3}\cos\frac{\pi}{2}\right)$$

$$= \frac{\pi}{2} - \frac{1}{3}$$

$$\int_{C_1+C_2+C_3} F \cdot dr = \int_{C_1} F \cdot dr + \int_{C_2} F \cdot dr + \int_{C_3} F \cdot dr$$

$$= 0 + \left(\frac{\pi}{2} - \frac{2}{3}\right) + \frac{1}{3}$$

$$= \frac{\pi}{2} - \frac{1}{3}$$

6(a) $F = \begin{bmatrix} -xy \\ y \\ yz \end{bmatrix}$

$C_1: r(t) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1-t \\ t \end{bmatrix}$ for $0 \le t \le 1$

$r'(t) = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

$F(t) = F(0, 1-t, t) = \begin{bmatrix} 0 \\ 1-t \\ t-t^2 \end{bmatrix}$

$$\int_{C_1} F \cdot dr = \int_0^1 F(t) \cdot r'(t)\, dt$$
$$= \int_0^1 \begin{bmatrix} 0 \\ 1-t \\ t-t^2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} dt$$
$$= \int_0^1 (t-1) + (t-t^2)\, dt$$
$$= \int_0^1 2t - t^2 - 1\, dt$$
$$= \left[t^2 - \tfrac{1}{3}t^3 - t\right]_0^1 = -\frac{1}{3}$$

$C_2: r(t) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ 1-t \end{bmatrix}$

$r'(t) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$F(t) = F(t, 0, 1-t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\int_{C_2} F \cdot dr = \int_0^1 F(t) \cdot r'(t)\, dt = 0$$

$C_3: r(t) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1-t \\ t \\ 0 \end{bmatrix}$ for $0 \le t \le 1$

$r'(t) = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$F(t) = F(1-t, t, 0) = \begin{bmatrix} t^2-t \\ t \\ 0 \end{bmatrix}$

$$\int_{C_3} F \cdot dr = \int_0^1 F(t) \cdot r'(t)\, dt$$
$$= \int_0^1 \begin{bmatrix} t^2-t \\ t \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} dt$$
$$= \int_0^1 t - t^2 + t\, dt$$
$$= \int_0^1 2t - t^2\, dt$$
$$= \left[t^2 - \tfrac{1}{3}t^3\right]_0^1$$
$$= \frac{2}{3}$$

6 (a) $\oint_C F \cdot dr = \int_{C_1} F \cdot dr + \int_{C_2} F \cdot dr + \int_{C_3} F \cdot dr$

$$= -\frac{1}{3} + 0 + \frac{2}{3}$$

$$= \frac{1}{3}$$

(b) $\text{curl } F = \nabla \times F$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -xy & y & yz \end{vmatrix}$$

$$= \begin{bmatrix} z - 0 \\ -(0-0) \\ 0 - (-x) \end{bmatrix}$$

$$= \begin{bmatrix} z \\ 0 \\ x \end{bmatrix}$$

(c). $x+y+z = 1$

$\Rightarrow$ $(0,0,1)$, $(1,0,0)$ and $(0,1,0)$ lies on the plane of interest.

$\Rightarrow$ $S$ is the triangular surface area bounded by $C_1$, $C_2$ and $C_3$

By Stokes Theorem

$$\iint_S \text{curl } F \cdot dS = \int_{C_1 + C_2 + C_3} F \cdot dr$$

$$= \int_{C_1} F \cdot dr + \int_{C_2} F \cdot dr + \int_{C_3} F \cdot dr$$

$$= -\frac{1}{3} + 0 + \frac{2}{3}$$

$$= \frac{1}{3}$$